Joy Schmus

jeschmus@gmail.com | (714) 815-1537 | www.linkedin.com/in/joyschmus

EDUCATION

Bachelor of Science in Engineering, GPA: 4.0 — Expected Graduation: May 2027
Biola University — La Mirada, CA
- Stewart Science Honors College · IEEE Secretary · Nancy Lorraine Jensen Memorial Scholarship Recipient

Semester Abroad — Spring 2026
Scholarship & Christianity in Oxford (SCIO) — Oxford, UK
- Completed tutorial-based engineering coursework emphasizing independent study and applied problem-solving
- Studied feedback control, system modeling, stability analysis, and state-space methods
- Practiced autonomous navigation, path planning, Bayesian filtering, and computer vision fundamentals

ENGINEERING PROJECTS & RESEARCH

Assistive Technology Senior Capstone Project — Fall 2026-Spring 2027
Biola University — La Mirada, CA
- Research control architectures and electromechanical design approaches for a wearable device assisting hand grasping and release with emphasis on portability, user safety, and reliable operation
- Evaluate actuator, motion-transmission, and sensing options, considering force requirements, degrees of freedom, and system response

Consultant for Drone Swarm Senior Capstone Project — Fall 2026-Spring 2027
Biola University — La Mirada, CA
- Evaluate hardware and communication requirements for a multi-drone autonomous light-show system using Pixhawk flight controllers and Skybrush software
- Assess compatibility of flight controllers, companion computers, telemetry radios, and positioning infrastructure to support reliable system operation

Quadcopter Control Systems Research — Spring 2026
Scholarship & Christianity in Oxford (SCIO) — Oxford, UK
- Modeled and simulated a cascading PID control for quadcopter flight using MATLAB, Simulink, and Simscape
- Developed and analyzed analog PID controllers using op-amp circuit models, comparing performance, stability, and disturbance response with digital PID implementations

Differential Drive Robot — Fall 2025
Biola University — La Mirada, CA
- Modeled and simulated rover kinematics and developed PI controllers using MATLAB, Simulink, and Stateflow
- Implemented path-planning and autonomous navigation algorithms using sensor feedback and waypoint logic
- Characterized motor behavior experimentally, integrated test data into control-system models, and diagnosed errors in localization, actuation, and feedback

ENGINEERING EXPERIENCE

Mechatronics Teaching and Lab Assistant — August 2026–Present
Biola University — La Mirada, CA
- Support hands-on mechatronics coursework covering PID control, motor sizing, system modeling, embedded systems, and communication protocols

Hardware Engineering Intern — May-August 2026
Palomar Products, Inc. — Rancho Santa Margarita, CA
- Designed and built an Aerial Refueling Amplifier test box, selecting components and developing wiring diagrams from MIL-SPEC requirements
- Characterized amplifier operation and signal levels using oscilloscopes, DMMs, and an audio analyzer to establish hardware interface requirements

Manufacturing Engineering Intern — June-August 2025
Kirkhill Inc. — Brea, CA
- Piloted cryogenic mold-cleaning experiments and presented findings that secured approval for implementation
- Performed root cause analysis to resolve defects on 4 rubber parts while updating process documentation for 14+ parts to standardize inspection and manufacturing procedures
- Designed SolidWorks inspection fixtures to improve dimensional verification and consistency

TECHNICAL SKILLS

MATLAB · Simulink · Simscape · Stateflow · C++ · Arduino · Digital Control Systems · PID Control · Feedback Control · System Modeling · Dynamic System Modeling · SolidWorks · Probabilistic Motion and Sensor Models